Week 8 Assignment
Contingency Tables and Chi-Square Tests

Course: H524 Introduction to Biostatistics
Total Points: 100 points
Submission: Canvas quiz

Instructions

This assignment has three parts:

- **Part A:** Multiple choice questions (70 points)
- **Part B:** R computational problems (30 points) - You will need to write and run R code
- **Part C:** Group project check-in (completion only, ungraded)

Submit your answers via the Canvas quiz.

1 Part A: Multiple Choice (70 points)

Contingency Table Basics (10 points)

A contingency table displays the relationship between:

A. One categorical variable

B. Two continuous variables

C. Two categorical variables

D. One categorical and one continuous variable

Expected Frequencies (10 points)

In a chi-square test, the expected frequency for a cell is calculated as:

A. (Row total × Column total) / Grand total

B. (Row total + Column total) / 2

C. Grand total / Number of cells

D. Observed frequency / Total sample size

Chi-Square Test Assumptions (10 points)

Which of the following is a requirement for conducting a chi-square test?

A. All expected cell frequencies must be at least 10

B. All expected cell frequencies should be at least 5

C. The sample must be normally distributed

D. Variables must be continuous

Degrees of Freedom (10 points)

For a 3×4 contingency table (3 rows, 4 columns), what are the degrees of freedom for the chi-square test?

A. 7

B. 6

C. 12

D. 11

Interpreting Chi-Square Results (10 points)

A chi-square test yields $p = 0.032$. At $\alpha = 0.05$, what can we conclude?

A. There is no association between the variables

B. There is a significant association between the variables

C. The variables are normally distributed

D. The test assumptions are violated

2×2 Table Setup (10 points)

In a case-control study examining smoking and lung cancer, which variable should form the rows of a 2×2 table?

A. Smoking status (exposed/unexposed)

B. Lung cancer status (case/control)

C. Either variable can form rows

D. Neither - case-control studies cannot use 2×2 tables

Fisher's Exact Test (10 points)

When should Fisher's exact test be used instead of the chi-square test?

A. When sample size is very large ($n > 1000$)

B. When expected cell frequencies are small (less than 5 in one or more cells)

C. When variables are continuous

D. When there are more than two categories

2 Part B: R Computational Problems (30 points)

Note: Some problems will require you to use R to compute the answer.

Question 8: Fisher's Exact Test in R (15 points)

A small clinical trial tests a new treatment for a rare disease with the following results:

	Improved	**Not Improved**
Treatment	7	3
Control	2	8

Use R to perform Fisher's exact test on this data. What is the two-sided p-value?

A. 0.0032

B. 0.0698

C. 0.1524

D. 0.5000

Question 9: Chi-Square Test in R (15 points)

A cohort study follows 500 smokers and 500 non-smokers for 10 years to assess heart disease incidence:

	Heart Disease	**No Heart Disease**
Smoker	75	425
Non-smoker	35	465

Use R to perform a chi-square test on this data (without continuity correction). What is the chi-square test statistic?

A. 8.234

B. 12.156

C. 16.343

D. 21.450

3 Part C: Group Project Check-In (Completion Only, Ungraded)

Progress Update Check-In

GROUP LEADER SUBMITS a brief progress update answering:

1. **Has your group met at least once since Week 7?** (Yes/No)
2. **Have you loaded and explored your dataset in R?** (Yes/No)
3. **Have you consulted at least one AI tool about your data?** (Yes/No - if yes, which one(s)?)
4. **Any blockers or questions?** (Optional)

Reminder: This Wednesday (12:00-1:20pm) is Office Hours Day - drop by classroom/office if your group needs help!

Submit as: Text entry (3-5 sentences).

Summary

- Part A (Multiple Choice): 70 points
- Part B (R Computational): 30 points
- Part C (Group Check-In): Completion only (ungraded)
- **Total: 100 points**